



SPACEMIT
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K3 CoM260 Kit

60 TOPS AI Compute System-on-Module (SoM)

Datasheet

2026/07/16

V1.3

Revision History

The revision history below records the updates made to this document.

Version	Date	Description
V1.3	2026.07.16	Updated Preface
V1.2	2026.07.10	Updated power-supply requirements and note
V1.1	2026.07.09	Updated wording and formatting
V1.0	2026.04.30	Initial release

Preface

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About This Document

This document describes the basic specifications and hardware features of the K3-CoM260. It is intended to help developers quickly understand the module specifications and carry out K3-CoM260 application and product development more accurately and efficiently.

This document is primarily intended for:

- Product managers
- Hardware engineers
- Other related engineering personnel

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1. Product Overview

1.1. Introduction

The K3-CoM260 is the first gold-finger package product launched by SpacemiT. It integrates the K3 RISC-V AI CPU, LPDDR5, GPYH, and other passive components.

The K3-CoM260 supports multiple operating systems and development platforms, including SpacemiT's native Bianbu OS, Ubuntu, Linux, and OpenHarmony. Together with a full-stack AI development toolkit, it can meet a wide range of AI development needs.

The K3-CoM260 provides a rich set of interfaces, including MIPI-DSI, MIPI-CSI, DP 1.2, SDIO 3.0, eSPI, SPI, I2S, I2C, CAN-FD, PWM, UART, USB 2.0, USB 3.0, PCIe 3.0, and GMAC. With its high performance and extensive I/O resources, the K3-CoM260 is a strong choice for RISC-V edge AI applications and product development.

1.2. Key Features

- **Outstanding CPU performance**

8 high-performance X100 compute cores combined with 8 A100 AI cores deliver more than 130 KDMIPS of CPU computing power. The X100 achieves SPECint2006 > 9.0/GHz per core, with a maximum frequency of 2.4 GHz.

- **Heterogeneous integrated AI computing**

Delivers 60 TOPS of AI computing power. Its ultra-wide parallel AI cores provide powerful general-purpose AI acceleration, enabling rapid integration with mainstream AI ecosystems. Local inference performance for 7B large models exceeds 10 tokens/s.

- **Latest RISC-V architecture with massive parallel computing capability**

The A100 supports RVV 1.0 parallel computing up to 1024 bits and complies with the latest RVA23 Profile and Vector Crypto standards.

- **Rich I/O expansion interfaces**

Integrates multiple high-speed expansion interfaces to flexibly support a wide range of expansion needs, including 8 PCIe lanes, 3 USB 3.0 ports, and 1 GMAC interface.

- **Comprehensive hardware virtualization**

Supports comprehensive hardware virtualization with RV Hypervisor 1.0, RV AIA, and RV IOMMU extensions, covering CPU, memory, interrupts, and I/O.

- **Advanced security protection**

Supports M/S/U processor privilege levels, hardware protection against attacks such as Spectre and RowHammer, and hardware security technologies compliant with SM2, SM3, and SM4.

- **Industrial-grade compliance**

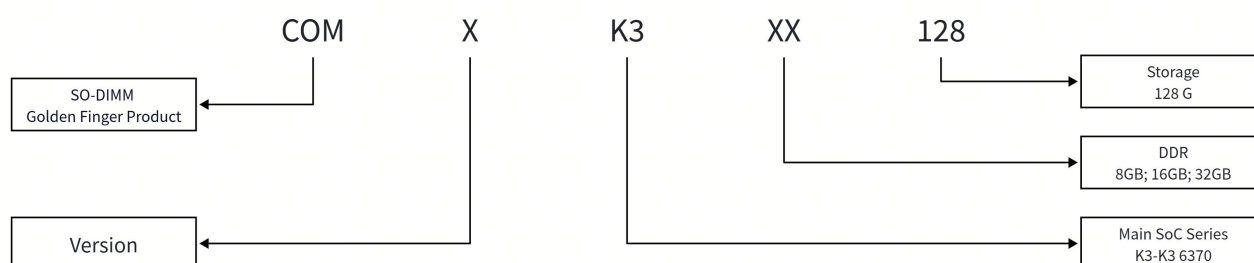
The CPU cores, multi-level cache, and SRAM support ECC design, enabling stable and reliable sustained compute performance over an ambient temperature range of -40°C ~ 85°C and meeting the stringent requirements of industrial applications.

1.3. Applications

- AI edge computing
- Integrated large-model inference systems
- Intelligent robots
- Product block diagrams

1.4. Ordering Information

The figure below shows the naming convention for **the ordering part number**, which consists of the following sections.

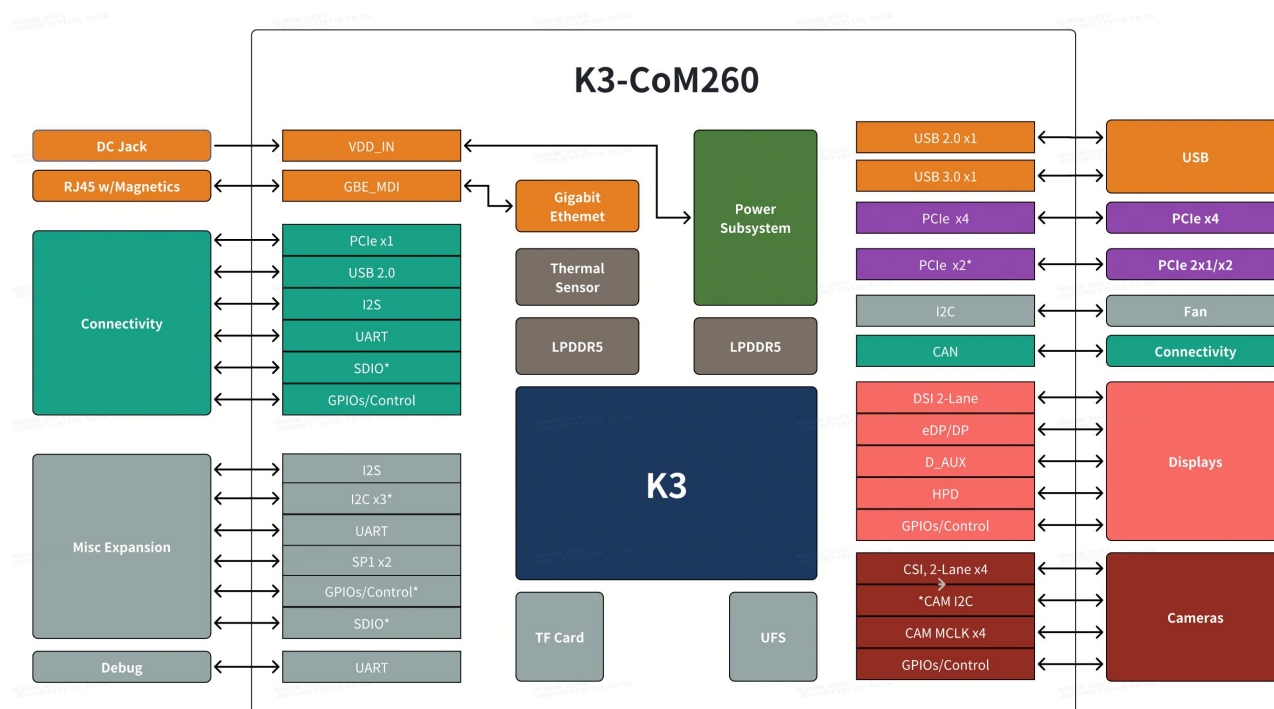


The table below lists the specific part numbers, main SoC model, and corresponding DDR capacity.

Part Number	Main SoC	DDR Capacity
COM3K308128	K3	8GB
COM3K316128	K3	16GB
COM3K332128	K3	32GB

2. Block Diagram

The block diagram of the K3-CoM260 is shown below.



Note: For detailed technical parameters of each module, refer to the corresponding [K3 Datasheet](#).

2.1. Interface Definitions

The tables below list the main interfaces of the K3-CoM260 and their related characteristics.

2.1.1. MIPI_DSI

Signal Name	Type	Description
MIPI_DSI1_LANE _x _DP[1:0]	O	DSI1 DATA LANE $\pm$
MIPI_DSI1_LANE _x _DN _x [1:0]	O	DSI1 DATA LANE $\pm$
MIPI_DSI1_CLKP	O	DSI1 Clock LANE $\pm$
MIPI_DSI1_CLKN	O	DSI1 Clock LANE $\pm$

2.1.2. MIPI_CSI

Signal Name	Type	Description
MIPI_CSI0_DP _x [1:0]	I	CSI0 DATA LANE $\pm$
MIPI_CSI0_DN _x [1:0]	I	CSI0 DATA LANE $\pm$
MIPI_CSI1_DP _x [3:0]	I	CSI1 DATA LANE $\pm$

Signal Name	Type	Description
MIPI_CSI1_DNx[3:0]	I	CSI1 DATA LANE $\pm$
MIPI_CSI2_DPx[1:0]	I	CSI2 DATA LANE $\pm$
MIPI_CSI2_DNx[1:0]	I	CSI2 DATA LANE $\pm$
MIPI_CSI3_DxP[1:0]	I	CSI3 DATA LANE $\pm$
MIPI_CSI3_DxN[1:0]	I	CSI3 DATA LANE $\pm$
MIPI_CSIx_CLKN[3:0]	O	CSI Clock LANE $\pm$
MIPI_CSIx_CLKP[3:0]	O	CSI Clock LANE $\pm$

2.1.3. DP 1.2

Signal Name	Type	Description
EDP1_TXxN[3:0]	O	EDP1 DATA LANE $\pm$
EDP1_TXxP[3:0]	O	
DP1_AUXN	O	EDP1 Auxiliary LANE $\pm$
DP1_AUXP	O	

2.1.4. eMMC

Signal Name	Type	Description
EMMC_Dx[7:0]	I/O	eMMC DATA
EMMC_CLK	I/O	eMMC Clock
EMMC_CMD	I/O	eMMC command
EMMC_DS	I/O	eMMC data strobe

2.1.5. SDIO 3.0

Signal Name	Type	Description
MMC1_DATx[3:0]	I/O	SD Card DATA
MMC1_CLK	I/O	SD Card Clock
MMC1_CMD	I/O	SD Card Command
MMC2_DATx[3:0]	I/O	SDIO DATA
MMC2_CLK	I/O	SDIO Clock
MMC2_CMD	I/O	SDIO Command

2.1.6. ESP I/O

Signal Name	Type	Description
ESPI0_Dx[3:0]	I/O	Four data lines for command, address, and data transfer. Operates in single-, dual-, or quad-I/O modes depending on configuration.
ESPI0_CS0	I	Chip Select, active low. Used by the master to select a specific slave device.
ESPI0_CLK	I	eSPI clock signal generated by the master. Synchronizes data transfer between master and slave.

2.1.7. SPI

Signal Name	Type	Description
SPix[2:0]_SCLK	I/O	Synchronous Serial Port Clock 0/2The serial bit clock can be configured as an output (master mode operation) or an input (slave mode operation).
SPix[2:0]_FRM	I/O	Synchronous Serial Port Frame 0/2The serial frame sync can be configured as an output (master mode operation) or an input (slave mode operation).
SPix[2:0]_TXD	O	Synchronous Serial Port Transmit Data 0/2Serial data driven out synchronously with the bit clock.
SPix[2:0]_RXD	I	Synchronous Serial Port Receive Data 0/2Serial data latched using the bit clock.
R.SPI0_SCLK	I/O	RT24 Synchronous Serial Port Clock 0/2The serial bit clock can be configured as an output (master mode operation) or an input (slave mode operation).
R.SPI0_FRM	I/O	RT24 Synchronous Serial Port Frame 0/2The serial frame sync can be configured as an output (master mode operation) or an input (slave mode operation).
R.SPI0_TXD	O	RT24 Synchronous Serial Port Transmit Data 0/2Serial data driven out synchronously with the bit clock.
R.SPI0_RXD	I	RT24 Synchronous Serial Port Receive Data 0/2Serial data latched using the bit clock.

2.1.8. I2S

Signal Name	Type	Description
I2S0_SCLK	I/O	I2S Continues Serial Clock

Signal Name	Type	Description
I2S0_LRCK	I/O	I2S Left-Right Clock
I2S0_TXD	O	I2S Serial Transmit Data
I2S0_RXD	I	I2S Serial Receive Data
I2Sx[5:2]_SCLK	I/O	I2S Continues Serial Clock
I2Sx[5:2]_LRCK	I/O	I2S Left-Right Clock
I2Sx[5:2]_TXD	O	I2S Serial Transmit Data
I2Sx[5:2]_RXD	I	I2S Serial Receive Data

2.1.9. I2C

Signal Name	Type	Description
I2Cx[6:0]_SDA	I/O	I2C Serial Data
I2Cx[6:0]_SCL	I/O	I2C Serial Clock
R.I2Cx[1:0]_SDA	I/O	RT24 domain I2C Serial Data
R.I2Cx[1:0]_SCL	I/O	RT24 domain I2C Serial Clock

2.1.10. CAN-FD

Signal Name	Type	Description
CANx[4:0]_TX	O	CAN Transmit Data
CANx[4:0]_RX	I	CAN Receive Data
R.CAN0_TX	O	RT24 domain CAN Transmit Data
R.CAN0_RX	I	RT24 domain CAN Receive Data
R.CAN2_TX	O	RT24 domain CAN Transmit Data
R.CAN2_RX	I	RT24 domain CAN Receive Data

2.1.11. FAN.

Signal Name	Type	Description
PWM	O	FAN Pulse-width modulation
TACH	I	FAN Tachometer

2.1.12. UART

Signal Name	Type	Description
UART0_TXD	O	DEBUG UART Transmit Data
UART0_RXD	I	DEBUG UART Receive Data
UART1_TXD	O	UART1 Transmit Data
UART1_RXD	I	UART1 Receive Data
UART1_CTS_N	I	UART1 Clear-to-Send
UART1_RTS_N	O	UART1 Request-to-Send
UARTx[10:3]_TXD	O	UART Transmit Data
UARTx[10:3]_RXD	I	UART Receive Data
UARTx[10:3]_CTS_N	I	UART Clear-to-Send
UARTx[10:3]_RTS_N	O	UART Request-to-Send
R.UART0_TXD	O	RT24 domain UART Transmit Data
R.UART0_RXD	I	RT24 domain UART Receive Data
R.UARTx[5:2]_TXD	O	RT24 domain UART Transmit Data
R.UARTx[5:2]_RXD	I	RT24 domain UART Receive Data

2.1.13. GPIO

Signal Name	Type	Description
GPIO[1]/[4]	I/O	3.3V GPIO domain
GPIO[2]/[3]/[5]	I/O	1.8V GPIO domain

2.1.14. USB 2.0

Signal Name	Type	Description
USB20_A_DRD_USB_M	I/O	DOWNLOAD USB D $\pm$, usually for USB 2.0
USB20_A_DRD_USB_P	I/O	

2.1.15. USB 3.0

Signal Name	Type	Description
USB20_A_DRD_USB_M	I/O	DOWNLOAD USB D $\pm$, usually for Download
USB20_A_DRD_USB_P	I/O	
USB_DRD_RXxN[1:0]	I/O	USB 3.0 Receive Data $\pm$ for Download

Signal Name	Type	Description
USB_DRD_RXxP[1:0]	I/O	
USB_DRD_TXxN[1:0]	I/O	USB 3.0 Transmit Data $\pm$ for Download
USB_DRD_TXxP[1:0]	I/O	

2.1.16. PCIe 3.0

Signal Name	Type	Description
PCIE_RXxN[1:0]	I/O	PCIe Receive Data $\pm$, usually for PCIeB, PCIeC
PCIE_RXxP[1:0]	I/O	
PCIE_TXxN[1:0]	I/O	PCIe Transmit Data $\pm$, usually for PCIeB, PCIeC
PCIE_TXxP[1:0]	I/O	
PCIE_REFCLK_N	I/O	PCIe Clock LANE $\pm$, usually for PCIeA, PCIeB, PCIeC
PCIE_REFCLK_P	I/O	

2.1.17. GMAC

Signal Name	Type	Description
GMAC1_RXDV	I	GMAC1 Receive Data Valid
GMAC1_RX_Dx[3:0]	I	GMAC1 Receive Data
GMAC1_RX_CLK	I	GMAC1 Receive Clock
GMAC1_TX_Dx[3:0]	O	GMAC1 Transmit Data
GMAC1_TX_EN	O	GMAC1 Transmit Data Enable
GMAC1_MDC	I/O	GMAC1 Configuration interface clock
GMAC1_MDIO	I/O	GMAC1 Configuration interface I/O
GMAC1_INT_N	I/O	GMAC1 Interrupt

3. Pin Definitions

3.1. Physical Pinout

The figure below shows the physical view of the K3-CoM260 module.



3.2. Pin Package

This section describes the pin package information of the K3-CoM260 in detail.

K3-CoM260 uses the following package specifications:

- Interface: gold finger (260-pin SODIMM)
- Dimensions: 69.6 mm × 45.0 mm × 1.2 mm
- Pin pitch: 0.5 mm

The table below shows the pin layout of the K3-CoM260.

Pin name	Pin NO.	Pin NO.	Pin Name
GND	1	2	GND
MIPI_CSI0_DN0	3	4	MIPI_CSI1_DN0
MIPI_CSI0_DP0	5	6	MIPI_CSI1_DP0

Pin name	Pin NO.	Pin NO.	Pin Name
GND	7	8	GND
MIPI_CSI0_CLKN	9	10	MIPI_CSI1_CLKN
MIPI_CSI0_CLKP	11	12	MIPI_CSI1_CLKP
GND	13	14	GND
MIPI_CSI0_DN1	15	16	MIPI_CSI1_DN1
MIPI_CSI0_DP1	17	18	MIPI_CSI1_DP1
GND	19	20	GND
MIPI_CSI2_DN2	21	22	MIPI_CSI2_DN0
MIPI_CSI2_DP2	23	24	MIPI_CSI2_DP0
GND	25	26	GND
MIPI_CSI3_CLKN	27	28	MIPI_CSI2_CLKN
MIPI_CSI3_CLKP	29	30	MIPI_CSI2_CLKP
GND	31	32	GND
MIPI_CSI2_DN3	33	34	MIPI_CSI2_DN1
MIPI_CSI2_DP3	35	36	MIPI_CSI2_DP1
GND	37	38	GND
USB_DRD_RX1N	39	40	PCIE4/USB3-D_RX0N
USB_DRD_RX1P	41	42	PCIE4/USB3-D_RX0P
GND	43	44	GND
USB_DRD_TX1N	45	46	PCIE4/USB3-D_TX0N
USB_DRD_TX1P	47	48	PCIE4/USB3-D_TX0P
GND	49	50	GND
USB_DRD_RX2N	51	52	PCIE4_REFCLK_N
USB_DRD_RX2P	53	54	PCIE4_REFCLK_P
GND	55	56	GND
USB_DRD_TX2N	57	58	PCIE3/USB3-C_RX0N
USB_DRD_TX2P	59	60	PCIE3/USB3-C_RX0P
GND	61	62	GND
EDP1_TX0N	63	64	PCIE3/USB3-C_TX0N
EDP1_TX0P	65	66	PCIE3/USB3-C_TX0P
GND	67	68	GND

Pin name	Pin NO.	Pin NO.	Pin Name
EDP1_TX1N	69	70	MIPI_DSI1_D0N
EDP1_TX1P	71	72	MIPI_DSI1_D0P
GND	73	74	GND
EDP1_TX2N	75	76	MIPI_DSI1_CLKN
EDP1_TX2P	77	78	MIPI_DSI1_CLKP
GND	79	80	GND
EDP1_TX3N	81	82	MIPI_DSI1_D1N
EDP1_TX3P	83	84	MIPI_DSI1_D1P
GND	85	86	GND
GPIO[3]_58 { GMAC2_PPS / R.UART2_RXD / R.CAN0_TXD / PCIEC_PERSTN / R.I2C0_SDA / PWM16 }	87	88	GPIO[3]_74 { R.GMAC3_CLK_REF / CLK_CAMCK2 / ESPI0_RESETN / VCXO_REQ / USB30H-1_DRV / R.I2C0_SCL }
GPIO[5]_104 { SSP0_TXD / SSP2_TXD / USB30H-1_DRV / CAN3_RXD / PCIED_PWRDET }	89	90	GPIO[3]_67 { R.GMAC3_TX_CLK / R.GPIO[22] / R.SSP1_FRM / CLK_CAMCK4 / PCIEC_PWRDET / R.PWM3 }
GPIO[5]_106 { SSP0_SCLK / SSP2_SCLK / R.I2C1_SDA / I2C3_SDA / PCIED_WAKEN / PWM18 }	91	92	GPIO[3]_53 { GMAC2_TX_EN / UART3_CTSN / SSP0_TXD / PCIEA_EINT / PWM11 }
GPIO[5]_105 { SSP0_RXD / SSP2_RXD / R.I2C1_SCL / I2C3_SCL / PCIED_PERSTN / PWM17 }	93	94	GPIO[1]_02 { GMAC0_RX_D1 / SSPA5_TXD / PWM2 / ESPI0_D2 / I2C1_SCL }
GPIO[5]_107 { SSP0_FRM / SSP2_FRM / R.CAN4_TXD / USB30-0_DIR / PCIED_CLKREQN / PWM19 }	95	96	GPIO[3]_72 { R.GMAC3_MDIO / SSPA4_RXD / ESPI0_CS / E/DP1_HPD / DSI0_TE }
GPIO[3]_70 { R.GMAC3_TX_EN / SSPA4_FRM / ESPI0_D2 / IR1_RX / MNCLK_OUT1 / SSP3_SCLK }	97	98	EDP1_AUXN
GPIO[3]_48 { GMAC2_TX_D0 / UART6_TXD / CAN1_RXD / PCIEA_AUXEN / I2C0_SCL / PWM6 }	99	100	EDP1_AUXP
GPIO[3]_49 { GMAC2_TX_D1 / UART6_RXD / CAN1_TXD / PCIEA_PWRDET / I2C0_SDA / PWM7 }	101	102	GND
GPIO[3]_51 { GMAC2_TX_D2 /	103	104	GPIO[3]_60 { R.GMAC3_RX_D0 /

Pin name	Pin NO.	Pin NO.	Pin Name
UART6_RTS / CAN2_RXD / PCIEA_ATNLED / I2C4_SDA / PWM9 }			R.UART5_RXD / R.SSP0_TXD / PCIEC_CLKREQN / R.I2C1_SDA / PWM18 }
GPIO[3]_50 { GMAC2_TX_CLK / UART6_CTS / CAN2_TXD / PCIEA_MRLN / I2C4_SCL / PWM8 }	105	106	GPIO[3]_62 { R.GMAC3_RX_CLK / R.SSP0_SCLK / PCIEC_ATTEN / I2C6_SDA }
GND	107	108	GPIO[3]_61 { R.GMAC3_RX_D1 / R.SSP0_RXD / PCIEC_PRSENT2N / I2C6_SCL / PWM19 }
USB20_A_DRD_USB_M	109	110	GPIO[3]_63 { R.GMAC3_RX_D2 / R.GPIO[18] / R.SSP0_FRM / PCIEC_PWRCTN / I2C5_SCL }
USB20_A_DRD_USB_P	111	112	GPIO[3]_71 { R.GMAC3_MDC / SSPA4_TXD / ESPI0_D3 / R.IR0_RX / MNCLK_OUT2 / SSP3_FRM }
GND	113	114	GPIO[3]_45 { GMAC2_RX_CLK / UART10_RXD / CAN0_RXD / PCIEA_PRSENT2N / PWM3 }
USB20_B_USB_M	115	116	GPIO[3]_75 { R.GMAC3_PPS / CLK_CAMCK1 / ESPI0_ALERT / VCXO_OUT / USB30H-2_DRV / R.I2C0_SDA }
USB20_B_USB_P	117	118	GPIO[5]_122 { MMC2_DAT[3] / SSPA1_CLK / UART6_TXD / R.UART0_TXD / PCIEB_ATNLED }
GND	119	120	GPIO[3]_44 { GMAC2_RX_D1 / UART10_TXD / CAN0_TXD / PCIEA_CLKREQN / PWM2 }
USB20_HOST_M	121	122	GPIO[4]_85 { CLK_CAMCK3 / SSPA0_SYSCCLK / UART9_RXD / USB30-2_DRV / PCIEA_AUXEN }
USB20_HOST_P	123	124	GPIO[4]_77 { R.SSPA0_FRM / SSPA2_FRM / UART8_RXD / CAN0_RXD / PCIEE_WAKEN / I2C0_SDA }
GND	125	126	GPIO[3]_57 { GMAC2_CLK_REF / R.UART2_TXD / R.CAN0_RXD / EDPO_HPDP / R.I2C0_SCL / PWM15 }

Pin name	Pin NO.	Pin NO.	Pin Name
GPIO[3]_59 { R.GMAC3_RXDV / R.UART5_TXD / PCIEC_WAKEN / R.I2C1_SCL / PWM17 }	127	128	GPIO[3]_73 { R.GMAC3_INT_N / SSPA4_SYSCLK / ESPI0_CLK / R.IR1_RX / USB20_HOST_DRV }
GND	129	130	GPIO[5]_103 { / SSPA3_SYSCLK / USB20_HOST_DRV / CAN3_TXD / PCIED_AUXEN / I2C1_SDA }
PCIE0_RX0N	131	132	GND
PCIE0_RX0P	133	134	PCIE0_TX0N
GND	135	136	PCIE0_TX0P
PCIE0_RX1N	137	138	GND
PCIE0_RX1P	139	140	PCIE0_TX1N
GND	141	142	PCIE0_TX1P
GPIO[4]_87 { R.SSP0_RXD / R.ESPI0_D1 / UART4_RXD / CAN2_RXD / PCIEA_MRLN / PCIEB_PRSENT2N }	143	144	GND
GPIO[4]_86 { R.SSP0_TXD / R.ESPI0_D0 / UART4_TXD / CAN2_TXD / PCIEA_PWRDET / USB30-0_DIR }	145	146	GND
GND	147	148	PCIE1_TX0N
PCIE1_RX0N	149	150	PCIE1_TX0P
PCIE1_RX0P	151	152	GND
GND	153	154	PCIE1_TX1N
PCIE1_RX1N	155	156	PCIE1_TX1P
PCIE1_RX1P	157	158	GND
GND	159	160	PCIE0_REFCLK_N
PCIE2/USB3-B_RX0N	161	162	PCIE0_REFCLK_P
PCIE2/USB3-B_RX0P	163	164	GND
GND	165	166	PCIE2/USB3-B_TX0N
PCIE5_RX0N	167	168	PCIE2/USB3-B_TX0P
PCIE5_RX0P	169	170	GND
GND	171	172	PCIE5_TX0N
PCIE5_REFCLK_N	173	174	PCIE5_TX0P

Pin name	Pin NO.	Pin NO.	Pin Name
PCIE5_REFCLK_P	175	176	GND
GND	177	178	GPIO[3]_42 { GMAC2_RXDV / UART0_TXD / PCIEA_PERSTN / I2C0_SCL / PWM0 }
GPIO[4]_80 { R.SSPA0_SYSCLK / SSPA2_SYSCLK / R.UART4_TXD / CAN3_RXD / PCIEA_WAKEN / I2C2_SCL }	179	180	GPIO[4]_81 { SSP0_TXD / SSPA0_CLK / R.UART4_RXD / CAN3_TXD / PCIEA_CLKREQN / I2C2_SDA }
GPIO[4]_79 { R.SSPA0_RXD / SSPA2_RXD / UART8_RTS / PCIEA_PERSTN / I2C1_SDA }	181	182	GPIO[4]_78 { R.SSPA0_TXD / SSPA2_TXD / UART8_CTS / PCIEE_CLKREQN / I2C1_SCL }
GPIO[4]_76 { R.SSPA0_CLK / SSPA2_CLK / UART8_TXD / CAN0_TXD / PCIEE_PERSTN / I2C0_SCL }	183	184	PHY1_MDI0-
GPIO[1]_00 { GMAC0_RXDV / SSPA5_CLK / PWM0 / IR1_RX / ESPI0_D0 / I2C0_SCL }	185	186	PHY1_MDI0+
GPIO[1]_01 { GMAC0_RX_D0 / SSPA5_FRM / PWM1 / R.IR1_RX / ESPI0_D1 / I2C0_SDA }	187	188	PHY1_LED1/CFG_LDO0
GPIO[4]_82 { SSP0_TXD / SSPA0_FRM / UART9_CTSN / UART5_RXD / PCIEA_PRSENT2N / I2C3_SCL }	189	190	PHY1_MDI1-
GPIO[4]_83 { SSP0_SCLK / SSPA0_TXD / UART9_RTSN / UART5_TXD / PCIEA_ATTEN / I2C3_SDA }	191	192	PHY1_MDI1+
GPIO[5]_113 { SSP1_SCLK / SSPA0_TXD / R.GPIO[30] / PCIEB_PERSTN / }	193	194	PHY1_LED2/CFG_LDO1
GPIO[5]_114 { SSP1_FRM / SSPA0_RXD / R.GPIO[31] / PCIEB_WAKEN / }	195	196	PHY1_MDI2-
GPIO[5]_112 { SSP1_RXD / SSPA0_FRM / UCIE_DESDA / I2C4_SDA / USB30-3_DRV / R.PWM9 }	197	198	PHY1_MDI2+
GPIO[5]_111 { SSP1_TXD / SSPA0_CLK / UCIE_DESCL / I2C4_SCL / USB30-	199	200	GND

Pin name	Pin NO.	Pin NO.	Pin Name
0_INT / R.PWM8 }			
GND	201	202	PHY1_MDI3-
GPIO[5]_121 { UART1_TXD / I2C2_SDA / R.CAN3_RXD / CAN4_RXD / PCIEB_MRLN }	203	204	PHY1_MDI3+
GPIO[5]_120 { UART1_RXD / I2C2_SCL / R.CAN3_TXD / CAN4_TXD / PCIEB_PWRDET }	205	206	GPIO[5]_123 { MMC2_DAT[2] / SSPA1_FRM / UART6_RXD / R.UART0_RXD / PCIEB_PWRLED / }
GPIO[5]_118 { UART1_RTSN / USB30_DRD_DRV / R.GPIO[35] / PCIEB_PWRCTN }	207	208	FAN_TACH(1V8)
GPIO[5]_119 { UART1_CTSN / USB30-0_INT / PCIEB_AUXEN }	209	210	PMIC_32K_OUT(3V3)
GPIO[5]_124 { MMC2_DAT[1] / SSPA1_TXD / PCIED_PERSTN / E/DP0_HPD / PCIEB_EINT / }	211	212	GPIO[3]_56 { GMAC2_INT_N / UART3_TXD / SSP0_FRM / R.UART3_TXD / PWM14 }
GPIO[4]_93 { R.GPIO[25] / R.ESPI0_ALERT / UART0_TXD / ESPI0_D2 / I2C5_SCL / R.PWM4 }	213	214	GPIO[3]_69 { R.GMAC3_TX_D3 / SSPA4_CLK / ESPI0_D1 / E/DP1_HPD / DSI0_TE / SSP3_RXD }
GPIO[4]_94 { R.GPIO[26] / UART0_RXD / ESPI0_D3 / I2C5_SDA / R.PWM6 }	215	216	GPIO[5]_125 { MMC2_DAT[0] / SSPA1_RXD / PCIED_WAKEN / E/DP1_HPD / PCIEB_EINTEG }
NA	217	218	GPIO[5]_126 { MMC2_CMD / SSPA1_SYSCLK / PCIED_CLKREQN / I2C5_SCL / }
GPIO[1]_03 { GMAC0_RX_CLK / SSPA5_RXD / PWM3 / PCIED_PERSTN / ESPI0_D3 / I2C1_SDA }	219	220	GPIO[5]_101 { SSP3_SCLK / SSPA3_TXD / UART4_CTS / CAN4_RXD / PCIED_ATTN / MNCLK_OUT1 }
GPIO[1]_05 { GMAC0_RX_D3 / PWM5 / PCIED_CLKREQN / ESPI0_CLK / I2C2_SCL }	221	222	GPIO[5]_102 { SSP3_FRM / SSPA3_RXD / UART4_RTS / CAN4_TXD / PCIED_PWRCTN / I2C1_SCL }
GPIO[1]_12 { GMAC0_MDC / UART7_CTSN / CAN0_RXD / PCIEC_PERSTN / UART8_TXD / I2C4_SDA }	223	224	GPIO[5]_100 { SSP3_RXD / SSPA3_FRM / UART4_RXD / R.CAN2_RXD / PCIED_PRSENTN / CLK32K_OUT }

Pin name	Pin NO.	Pin NO.	Pin Name
GPIO[1]_14 { GMAC0_INT_N / UART7_RXD / PWM14 / PCIEC_CLKREQN / MNCLK_OUT1 / I2C6_SCL }	225	226	GPIO[5]_99 { SSP3_TXD / SSPA3_CLK / UART4_TXD / R.CAN2_TXD / CLK_CAMCK4 }
PCIE3_REFCLK_N	227	228	GPIO[5]_127 { MMC2_CLK / PCIED_PRSENT2N / I2C5_SDA / USB30-2_DRV }
PCIE3_REFCLK_P	229	230	FAN_PWM(1V8)
GND	231	232	GPIO[3]_54 { GMAC2_MDC / UART3_RTSN / SSP0_RXD / PCIEA_EINTEG / I2C1_SCL / PWM12 }
GPIO[4]_88 { R.SSP0_SCLK / R.ESPI0_D2 / R.UART3_TXD / PCIEB_PERSTN / PCIEA_ATNLED / CAN1_RXD }	233	234	GPIO[3]_55 { GMAC2_MDIO / UART3_RXD / SSP0_SCLK / R.UART3_RXD / I2C1_SDA / PWM13 }
VCC_RTC(5V)	235	236	PWR_SSP_SCLK { GPIO[122] / UART0_TXD }
POWER_EN(5V)	237	238	PWR_SSP_FRM { GPIO[123] / UART0_RXD }
PMIC_RST_OUTn(1V8)	239	240	GPIO[4]_84 { SSP0_FRM / SSPA0_RXD / UART9_TXD / USB30-1_DRV / PCIEA_PWRCTN / DSI0_TE }
GND	241	242	GND
GND	243	244	GND
GND	245	246	GND
GND	247	248	GND
GND	249	250	GND
VDD_IN(5V/20V)	251	252	VDD_IN(5V/20V)
VDD_IN(5V/20V)	253	254	VDD_IN(5V/20V)
VDD_IN(5V/20V)	255	256	VDD_IN(5V/20V)
VDD_IN(5V/20V)	257	258	VDD_IN(5V/20V)
VDD_IN(5V/20V)	259	260	VDD_IN(5V/20V)

The table below lists the pin type definitions of the K3-CoM260.

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
1	GND		G	GND	Digital core Ground	
3	MIPI_CSI0_DN0	AG39	I		MIPI_CSI0_DN0	CSI0 DATA0 LANEN
5	MIPI_CSI0_DP0	AG38	I		MIPI_CSI0_DP0	CSI0 DATA0 LANEP
7	GND		G		GND	Digital core Ground
9	MIPI_CSI0_CLKN	AF36	O		MIPI_CSI0_CLKN	CSI0 Clock LANEN
11	MIPI_CSI0_CLKP	AF37	O		MIPI_CSI0_CLKP	CSI0 Clock LANEP
13	GND		G		GND	Digital core Ground
15	MIPI_CSI0_DN1	AF40	I		MIPI_CSI0_DN1	CSI0 DATA1 LANEN
17	MIPI_CSI0_DP1	AF39	I		MIPI_CSI0_DP1	CSI0 DATA1 LANEP
19	GND		G	GND	Digital core Ground	
21	MIPI_CSI2_DN2	V39	I		MIPI_CSI2_DN2	CSI2 DATA2 LANEN
23	MIPI_CSI2_DP2	V40	I		MIPI_CSI2_DP2	CSI2 DATA2 LANEP
25	GND		G	GND	Digital core Ground	
27	MIPI_CSI3_CLKN	W38	O		MIPI_CSI3_CLKN	CSI3 Clock LANEN
29	MIPI_CSI3_CLKP	W39	O		MIPI_CSI3_CLKP	CSI3 Clock LANEP
31	GND		G	GND	Digital core Ground	
33	MIPI_CSI2_DN3	V36	I		MIPI_CSI2_DN3	CSI2 DATA3 LANEN
35	MIPI_CSI2_DP3	V37	I		MIPI_CSI2_DP3	CSI2 DATA3 LANEP
37	GND		G	GND	Digital core Ground	
39	USB_DRD_RX1N	AT12	I		USB_DRD_RX1N	USB3.0_DRD RX1 LANEN
41	USB_DRD_RX1P	AU12	I		USB_DRD_RX1P	USB3.0_DRD RX1 LANEP
43	GND		G	GND	Digital core Ground	
45	USB_DRD_TX1N	AY10	O		USB_DRD_TX1N	USB3.0_DRD TX1 LANEN
47	USB_DRD_TX1P	AW10	O		USB_DRD_TX1P	USB3.0_DRD TX1 LANEP
49	GND		G	GND	Digital core Ground	
51	USB_DRD_RX2N	AW11	I		USB_DRD_RX2N	USB3.0_DRD RX2 LANEN
53	USB_DRD_RX2P	AV11	I		USB_DRD_RX2P	USB3.0_DRD RX2 LANEP
55	GND		G	GND	Digital core Ground	
57	USB_DRD_TX2N	AW12	O		USB_DRD_TX2N	USB3.0_DRD TX2 LANEN

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
59	USB_DRD_TX2P	AY12	O		USB_DRD_TX2P	USB3.0_DRD TX2 LANEP
61	GND		G	GND	Digital core Ground	
63	EDP1_TX0N	AT18	O		DP1_TX0N	DP DATA0 LANEN
65	EDP1_TX0P	AU18	O		DP1_TX0P	DP DATA0 LANEP
67	GND		G	GND	Digital core Ground	
69	EDP1_TX1N	AW18	O		DP1_TX1N	DP DATA1 LANEN
71	EDP1_TX1P	AY18	O		DP1_TX1P	DP DATA1 LANEP
73	GND		G	GND	Digital core Ground	
75	EDP1_TX2N	AW19	O		DP1_TX2N	DP DATA2 LANEN
77	EDP1_TX2P	AV19	O		DP1_TX2P	DP DATA2 LANEP
79	GND		G	GND	Digital core Ground	
81	EDP1_TX3N	AW20	O		DP1_TX3N	DP DATA3 LANEN
83	EDP1_TX3P	AY20	O		DP1_TX3P	DP DATA3 LANEP
85	GND		G	GND	Digital core Ground	
87	GPIO[3]_58 {GMAC2_PPS/R.UART2_RXD /R.CAN0_TXD/PCIEC_PERSTN/R.I2C0_SDA/PWM16 }	M36	I	1.8V	USB0_VBUS_DET	General purpose I/O3 58
89	GPIO[5]_104 { SSP0_TXD / SSP2_TXD / USB30H-1_DRV / CAN3_RXD / PCIED_PWRDET / }	AW27	O	1.8V	SPI0_MOSI	SPI0_MOSI
91	GPIO[5]_106 { SSP0_SCLK / SSP2_SCLK / R.I2C1_SDA / I2C3_SDA / PCIED_WAKEN / PWM18 }	AP27	O	1.8V	SPI0_SCK	SPI0_SCK
93	GPIO[5]_105 { SSP0_RXD / SSP2_RXD / R.I2C1_SCL / I2C3_SCL / PCIED_PERSTN / PWM17 }	AR27	I	1.8V	SPI0_MISO	SPI0_MISO
95	GPIO[5]_107 { SSP0_FRM / SSP2_FRM / R.CAN4_TXD / USB30-0_DIR / PCIED_CLKREQN / PWM19 }	AY27	I	1.8V	SPI0_CS	SPI0_CS

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
97	GPIO[3]_70 { R.GMAC3_TX_EN / SSPA4_FRM / ESPIO_D2 / IR1_RX / MNCLK_OUT1 / SSP3_SCLK }	J40	I	1.8V	SPI0_CS1	General purpose I/O3 70
99	GPIO[3]_48 { GMAC2_TX_D0 / UART6_TXD / CAN1_RXD / PCIEA_AUXEN / I2C0_SCL / PWM6 }	J35	O	1.8V	UART6_TXD	UART6_TXD
101	GPIO[3]_49 { GMAC2_TX_D1 / UART6_RXD / CAN1_TXD / PCIEA_PWRDET / I2C0_SDA / PWM7 }	K35	I	1.8V	UART6_RXD	UART6_RXD
103	GPIO[3]_51 { GMAC2_TX_D2 / UART6_RTS / CAN2_RXD / PCIEA_ATNLED / I2C4_SDA / PWM9 }	M35	I	1.8V	UART6_RTS	UART6_RTS
105	GPIO[3]_50 { GMAC2_TX_CLK / UART6_CTS / CAN2_TXD / PCIEA_MRLN / I2C4_SCL / PWM8 }	L35	O	1.8V	UART6_CTS	UART6_RTS
107	GND		G	GND	Digital core Ground	
109	USB20_A_DRD_USB_M	AU14	I/O		USB20_A_DRD_USB_M	USB2.0_A_DRD D- differential data line
111	USB20_A_DRD_USB_P	AT14	I/O		USB20_A_DRD_USB_P	USB2.0_A_DRD D+ differential data line
113	GND		G	GND	Digital core Ground	
115	USB20_B_USB_M	C18	I/O		USB20_B_USB_M	USB2.0_B D- differential data line
117	USB20_B_USB_P	B18	I/O		USB20_B_USB_P	USB2.0_B D+ differential data line
119	GND		G	GND	Digital core Ground	
121	USB20_HOST_M	B22	I/O		USB20_HOST_M	USB2.0_HOST D- differential data line
123	USB20_HOST_P	C22	I/O		USB20_HOST_P	USB2.0_HOST D+

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
						differential data line
125	GND		G	GND	Digital core Ground	
127	GPIO[3]_59 { R.GMAC3_RXDV / R.UART5_TXD / PCIEC_WAKEN / R.I2C1_SCL / PWM17 }	J38	O	1.8V	PWR_LED_CTRL	General purpose I/O3 59
129	GND		G	GND	Digital core Ground	
131	PCIE0_RX0N	G22	I		PCIE0_RX0N	PCIE0 RX0 LANEN
133	PCIE0_RX0P	F22	I		PCIE0_RX0P	PCIE0 RX0 LANEP
135	GND		G	GND	Digital core Ground	
137	PCIE0_RX1N	E21	I		PCIE0_RX1N	PCIE0 RX1 LANEN
139	PCIE0_RX1P	D21	I		PCIE0_RX1P	PCIE0 RX1 LANEP
141	GND		G	GND	Digital core Ground	
143	GPIO[4]_87 { R.SSP0_RXD / R.ESPI0_D1 / UART4_RXD / CAN2_RXD / PCIEA_MRLN / PCIEB_PRSENT2N }	AU35	I	3.3V	CAN2_RXD	CAN2_RXD
145	GPIO[4]_86 { R.SSP0_TXD / R.ESPI0_D0 / UART4_TXD / CAN2_TXD / PCIEA_PWRDET / USB30-0_DIR }	AT35	O	3.3V	CAN2_TXD	CAN2_TXD
147	GND		G	GND	Digital core Ground	
149	PCIE1_RX0N	E19	I		PCIE1_RX0N	PCIE1 RX0 LANEN
151	PCIE1_RX0P	D19	I		PCIE1_RX0P	PCIE1 RX0 LANEP
153	GND		G	GND	Digital core Ground	
155	PCIE1_RX1N	F18	I		PCIE1_RX1N	PCIE1 RX1 LANEN
157	PCIE1_RX1P	G18	I		PCIE1_RX1P	PCIE1 RX1 LANEP
159	GND		G	GND	Digital core Ground	
161	PCIE2/USB3-B_RX0N	E17	I		PCIE2/USB3-B_RX0N	USB3.0_B RX0 LANEN
163	PCIE2/USB3-B_RX0P	D17	I		PCIE2/USB3-B_RX0P	USB3.0_B RX0 LANEP
165	GND		G	GND	Digital core Ground	
167	PCIE5_RX0N	E9	I		PCIE5_RX0N	PCIE5 RX0 LANEN

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
169	PCIE5_RX0P	D9	I		PCIE5_RX0P	PCIE5 RX0 LANEP
171	GND		G	GND	Digital core Ground	
173	PCIE5_REFCLK_N	B8	O		PCIE5_REFCLK_N	PCIE5 Clock LANEN
175	PCIE5_REFCLK_P	C8	O		PCIE5_REFCLK_P	PCIE5 Clock LANEP
177	GND		G	GND	Digital core Ground	
179	GPIO[4]_80 { R.SSPA0_SYSCLK / SSPA2_SYSCLK / R.UART4_TXD / CAN3_RXD / PCIEA_WAKEN / I2C2_SCL }	AR34	I	3.3V	PCleA_WAKEn	PCleA_WAKEn
181	GPIO[4]_79 { R.SSPA0_RXD / SSPA2_RXD / UART8_RTS / PCIEA_PERSTN / I2C1_SDA }	AP34	O	3.3V	PCleA_PERSTn	PCleA_PERSTn
183	GPIO[4]_76 { R.SSPA0_CLK / SSPA2_CLK / UART8_TXD / CAN0_TXD / PCIEE_PERSTN / I2C0_SCL }	AR33	O	3.3V	PCleE_PERSTn	PCleA_PERSTn
185	GPIO[1]_00 { GMAC0_RXDV / SSPA5_CLK / PWM0 / IR1_RX / ESPI0_D0 / I2C0_SCL }	AY32	O	3.3V	I2C0_SCL	I2C0_SCL
187	GPIO[1]_01 { GMAC0_RX_D0 / SSPA5_FRM / PWM1 / R.IR1_RX / ESPI0_D1 / I2C0_SDA }	AW32	I/O	3.3V	I2C0_SDA	I2C0_SDA
189	GPIO[4]_82 { SSP0_RXD / SSPA0_FRM / UART9_CTSN / UART5_RXD / PCIEA_PRSTN2N / I2C3_SCL }	AV34	O	3.3V	I2C3_SCL	I2C3_SCL
191	GPIO[4]_83 { SSP0_SCLK / SSPA0_TXD / UART9_RTSN / UART5_TXD / PCIEA_ATTN / I2C3_SDA }	AW34	I/O	3.3V	I2C3_SDA	I2C3_SDA
193	GPIO[5]_113 { SSP1_SCLK / SSPA0_TXD / R.GPIO[30] / PCIEB_PERSTN }	AY26	O	1.8V	I2S0_SDOUT	I2S0_SDOUT

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
195	GPIO[5]_114 { SSP1_FRM / SSPA0_RXD / R.GPIO[31] / PCIEB_WAKEN }	AP25	I	1.8V	I2S0_SDIN	I2S0_SDIN
197	GPIO[5]_112 { SSP1_RXD / SSPA0_FRM / UCIE_DESDA / I2C4_SDA / USB30-3_DRV / R.PWM9 }	AW26	I	1.8V	I2S0_LRCK	I2S0_LRCK
199	GPIO[5]_111 { SSP1_TXD / SSPA0_CLK / UCIE_DESCL / I2C4_SCL / USB30-0_INT / R.PWM8 }	AU26	O	1.8V	I2S0_CLK	I2S0_CLK
201	GND		G	GND	Digital core Ground	
203	GPIO[5]_121 { UART1_TXD / I2C2_SDA / R.CAN3_RXD / CAN4_RXD / PCIEB_MRLN }	AT24	O	1.8V	UART1_TXD	UART1_TXD
205	GPIO[5]_120 { UART1_RXD / I2C2_SCL / R.CAN3_TXD / CAN4_TXD / PCIEB_PWRDET / }	AR24	I	1.8V	UART1_RXD	UART1_TXD
207	GPIO[5]_118 { UART1_RTSTN / USB30_DRD_DRV / R.GPIO[35] / PCIEB_PWRCTN }	AW25	I	1.8V	UART1_RTSTn	UART1_RTSTn
209	GPIO[5]_119 { UART1_CTSTN / USB30-0_INT / PCIEB_AUXEN / }	AP24	O	1.8V	UART1_CTSTn	UART1_CTSTn
211	GPIO[5]_124 { MMC2_DAT[1] / SSPA1_TXD / PCIED_PERSTN / E/DP0_HPD / PCIEB_EINT }	AT23	I/O	1.8V	MMC2_DAT1	General purpose I/O5 124
213	GPIO[4]_93 { R.GPIO[25] / R.ESPI0_ALERT / UART0_TXD / ESPI0_D2 / I2C5_SCL / R.PWM4 }	AW37	O	3.3V	I2C5_SCL	I2C5_SCL
215	GPIO[4]_94 { R.GPIO[26] / UART0_RXD / ESPI0_D3 / I2C5_SDA / R.PWM6 }	AY37	I/O	3.3V	I2C5_SDA	I2C5_SDA

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
217	NA		I/O	3.3V	MODULE_ID	NA
219	GPIO[1]_03 { GMAC0_RX_CLK / SSPA5_RXD / PWM3 / PCIED_PERSTN / ESPI0_D3 / I2C1_SDA }	AU32	O	3.3V	PCleD_PERSTn	PCleD_PERSTn
221	GPIO[1]_05 { GMAC0_RX_D3 / PWM5 / PCIED_CLKREQN / ESPI0_CLK / I2C2_SCL }	AP32	I	3.3V	PCleD_CLKREQn	PCleD_CLKREQn
223	GPIO[1]_12 { GMAC0_MDC / UART7_CTSN / CAN0_RXD / PCIEC_PERSTN / UART8_TXD / I2C4_SDA }	AW30	O	3.3V	PCleC_PERSTn	PCleC_PERSTn
225	GPIO[1]_14 { GMAC0_INT_N / UART7_RXD / PWM14 / PCIEC_CLKREQN / MNCLK_OUT1 / I2C6_SCL }	AW29	I	3.3V	PCleC_CLKREQn	PCleC_CLKREQn
227	PCIE3_REFCLK_N	B12	O		PCIE3_REFCLK_N	PCIE3 Clock LANEN
229	PCIE3_REFCLK_P	C12	O		PCIE3_REFCLK_P	PCIE3 Clock LANEP
231	GND		G	GND	Digital core Ground	
233	GPIO[4]_88 { R.SSP0_SCLK / R.ESPI0_D2 / R.UART3_TXD / PCIEB_PERSTN / PCIEA_ATNLED / CAN1_RXD }	AV35	O	3.3V	SHUTDOWN_REQ	General purpose I/O4 88
235	VCC_RTC(5V)			1.85-5.5V	VCC_RTC	RTC power
237	POWER_EN(5V)		I	5V	POWER_EN	K3-CoM260 Power Enable
239	PMIC_RST_OUTn(1V8)		O	1.8V	PMIC_RST_OUTn	K3-CoM260 Reset
241	GND		G	G	GND	Digital core Ground
243	GND					
245	GND					
247	GND					
249	GND					
251	VDD_IN(5V/20V)		P	5V/9~19V	VDD_IN	K3-CoM260 Power

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
253	VDD_IN(5V/20V)					
255	VDD_IN(5V/20V)					
257	VDD_IN(5V/20V)					
259	VDD_IN(5V/20V)					
2	GND		G	GND	Digital core Ground	
4	MIPI_CSI1_DN0	AD37	I		MIPI_CSI1_DN0	CSI1 DATA0 LANEN
6	MIPI_CSI1_DP0	AD36	I		MIPI_CSI1_DP0	CSI1 DATA0 LANEP
8	GND		G		GND	Digital core Ground
10	MIPI_CSI1_CLKN	AB39	O		MIPI_CSI1_CLKN	CSI1 Clock LANEN
12	MIPI_CSI1_CLKP	AB40	O		MIPI_CSI1_CLKP	CSI1 Clock LANEP
14	GND		G		GND	Digital core Ground
16	MIPI_CSI1_DN1	AC39	I		MIPI_CSI1_DN1	CSI1 DATA1 LANEN
18	MIPI_CSI1_DP1	AC38	I		MIPI_CSI1_DP1	CSI1 DATA1 LANEP
20	GND		G	GND	Digital core Ground	
22	MIPI_CSI2_DN0	Y40	I		MIPI_CSI2_DN0	CSI2 DATA0 LANEN
24	MIPI_CSI2_DP0	Y39	I		MIPI_CSI2_DP0	CSI2 DATA0 LANEP
26	GND		G	GND	Digital core Ground	
28	MIPI_CSI2_CLKN	AB33	O		MIPI_CSI2_CLKN	CSI2 Clock LANEN
30	MIPI_CSI2_CLKP	AB34	O		MIPI_CSI2_CLKP	CSI2 Clock LANEP
32	GND		G	GND	Digital core Ground	
34	MIPI_CSI2_DN1	Y37	I		MIPI_CSI2_DN1	CSI2 DATA1 LANEN
36	MIPI_CSI2_DP1	Y36	I		MIPI_CSI2_DP1	CSI2 DATA1 LANEP
38	GND		G	GND	Digital core Ground	
40	PCIE4/USB3-D_RX0N	D13	I		PCIE4/USB3-D_RX0N	PCIE4 RX0 LANEN
42	PCIE4/USB3-D_RX0P	E13	I		PCIE4/USB3-D_RX0P	PCIE4 RX0 LANEP
44	GND		G	GND	Digital core Ground	
46	PCIE4/USB3-D_TX0N	A9	O		PCIE4/USB3-D_TX0N	PCIE4 TX0 LANEN
48	PCIE4/USB3-D_TX0P	B9	O		PCIE4/USB3-D_TX0P	PCIE4 TX0 LANEP
50	GND		G	GND	Digital core Ground	
52	PCIE4_REFCLK_N	C10	O		PCIE4_REFCLK_N	PCIE4 Clock LANEN

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
54	PCIE4_REFCLK_P	B10	O		PCIE4_REFCLK_P	PCIE4 Clock LANEP
56	GND		G	GND	Digital core Ground	
58	PCIE3/USB3-C_RX0N	D15	I		PCIE3/USB3-C_RX0N	PCIE3 RX0 LANEN
60	PCIE3/USB3-C_RX0P	E15	I		PCIE3/USB3-C_RX0P	PCIE3 RX0 LANEP
62	GND		G	GND	Digital core Ground	
64	PCIE3/USB3-C_TX0N	A11	O		PCIE3/USB3-C_TX0N	PCIE3 TX0 LANEN
66	PCIE3/USB3-C_TX0P	B11	O		PCIE3/USB3-C_TX0P	PCIE3 TX0 LANEP
68	GND		G	GND	Digital core Ground	
70	MIPI_DSI1_D0N	AK34	O		MIPI_DSI1_D0N	DSI 1 DATA0 LANEN
72	MIPI_DSI1_D0P	AK33	O		MIPI_DSI1_D0P	DSI 1 DATA0 LANEP
74	GND		G	GND	Digital core Ground	
76	MIPI_DSI1_CLKN	AJ38	O		MIPI_DSI1_CLKN	DSI 1 Clock LANEN
78	MIPI_DSI1_CLKP	AJ39	O		MIPI_DSI1_CLKP	DSI 1 Clock LANEP
80	GND		G	GND	Digital core Ground	
82	MIPI_DSI1_D1N	AH39	O		MIPI_DSI1_D1N	DSI 1 DATA1 LANEN
84	MIPI_DSI1_D1P	AH40	O		MIPI_DSI1_D1P	DSI 1 DATA1 LANEP
86	GND		G	GND	Digital core Ground	
88	GPIO[3]_74 { R.GMAC3_CLK_REF / CLK_CAMCK2 / ESPI0_RESETN / VCXO_REQ / USB30H-1_DRV / R.I2C0_SCL }	N40	O	1.8V	CLK_CAMCK2	CLK_CAMCK2
90	GPIO[3]_67 { R.GMAC3_TX_CLK / R.GPIO[22] / R.SSP1_FRM / CLK_CAMCK4 / PCIEC_PWRDET / R.PWM3 }	M39	O	1.8V	CLK_CAMCK4	CLK_CAMCK4
92	GPIO[3]_53 { GMAC2_TX_EN / UART3_CTSN / SSP0_TXD / PCIEA_EINT / PWM11 }	H36	O	1.8V	MIPI_CSI3_PWDN	General purpose I/O3 53
94	GPIO[1]_02 { GMAC0_RX_D1 / SSPA5_TXD / PWM2 / ESPI0_D2 / I2C1_SCL }	AV32	O	1.8V	MIPI_CSI1_PWDN	General purpose I/O1 02

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
96	GPIO[3]_72 { R.GMAC3_MDIO / SSPA4_RXD / ESPI0_CS / E/DP1_HPD / DSI0_TE }	L40	I	1.8V	e/DP1_HPD	e/DP1_HPD
98	EDP1_AUXN	AW17	I/O		DP1_AUXN	DP1 AUX LANEN
100	EDP1_AUXP	AV17	I/O		DP1_AUXP	DP1 AUX LANEP
102	GND		G	GND	Digital core Ground	
104	GPIO[3]_60 { R.GMAC3_RX_D0 / R.UART5_RXD / R.SSP0_TXD / PCIEC_CLKREQN / R.I2C1_SDA / PWM18 }	L38	O	1.8V	R-SPI0_MOSI	R.SPI0_MOSI
106	GPIO[3]_62 { R.GMAC3_RX_CLK / R.SSP0_SCLK / PCIEC_ATTN / I2C6_SDA }	N38	O	1.8V	R-SPI0_SCK	R.SPI0_SCK
108	GPIO[3]_61 { R.GMAC3_RX_D1 / R.SSP0_RXD / PCIEC_PRSENT2N / I2C6_SCL / PWM19 }	M38	I	1.8V	R-SPI0_MISO	R.SPI0_MISO
110	GPIO[3]_63 { R.GMAC3_RX_D2 / R.GPIO[18] / R.SSP0_FRM / PCIEC_PWRCTN / I2C5_SCL }	P38	I	1.8V	R-SPI0_CS	R.SPI0_CS
112	GPIO[3]_71 { R.GMAC3_MDC / SSPA4_TXD / ESPI0_D3 / R.IR0_RX / MNCLK_OUT2 / SSP3_FRM }	K40	I	1.8V	R-SPI0_CS1	General purpose I/O3 71
114	GPIO[3]_45 { GMAC2_RX_CLK / UART10_RXD / CAN0_RXD / PCIEA_PRSENT2N / PWM3 }	L34	O	1.8V	MIPI_CSI0_PWDN	General purpose I/O3 45
116	GPIO[3]_75 { R.GMAC3_PPS / CLK_CAMCK1 / ESPI0_ALERT / VCXO_OUT / USB30H-2_DRV /	P40	O	1.8V	CLK_CAMCK1	General purpose I/O3 75

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
	R.I2C0_SDA }					
118	GPIO[5]_122 { MMC2_DAT[3] / SSPA1_CLK / UART6_TXD / R.UART0_TXD / PCIEB_ATNLED }	AW24	I/O	1.8V	MMC2_DAT3	General purpose I/O5 122
120	GPIO[3]_44 { GMAC2_RX_D1 / UART10_TXD / CAN0_TXD / PCIEA_CLKREQN / PWM2 }	K34	O	1.8V	MIPI_CSI2_PWDN	General purpose I/O3 44
122	GPIO[4]_85 { CLK_CAMCK3 / SSPA0_SYSCCLK / UART9_RXD / USB30-2_DRV / PCIEA_AUXEN }	AY33	O	1.8V	CLK_CAMCK3	CLK_CAMCK3
124	GPIO[4]_77 { R.SSPA0_FRM / SSPA2_FRM / UART8_RXD / CAN0_RXD / PCIEE_WAKEN / I2C0_SDA }	AT33	I	3.3V	BT_M2_WAKE_AP	General purpose I/O4 77
126	GPIO[3]_57 { GMAC2_CLK_REF / R.UART2_TXD / R.CAN0_RXD / EDP0_HPD / R.I2C0_SCL / PWM15 }	L37	O	1.8V	BT_M2_EN	General purpose I/O3 57
128	GPIO[3]_73 { R.GMAC3_INT_N / SSPA4_SYSCCLK / ESPI0_CLK / R.IR1_RX / USB20_HOST_DRV }	M40	O	1.8V	W_DISABLE1_CTRL	General purpose I/O3 73
130	GPIO[5]_103 { SSPA3_SYSCCLK / USB20_HOST_DRV / CAN3_TXD / PCIED_AUXEN / I2C1_SDA }	AY28	O	1.8V	CAM_MUX_SEL	General purpose I/O5 103
132	GND		G	GND	Digital core Ground	
134	PCIE0_TX0N	A21	O		PCIE0_TX0N	PCIE0 TX0 LANEN
136	PCIE0_TX0P	B21	O		PCIE0_TX0P	PCIE0 TX0 LANEP
138	GND		G	GND	Digital core Ground	
140	PCIE0_TX1N	B19	O		PCIE0_TX1N	PCIE0 TX1 LANEN

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
142	PCIE0_TX1P	A19	O		PCIE0_TX1P	PCIE0 TX1 LANEP
144	GND		G	GND	Digital core Ground	
146	GND		G	GND	Digital core Ground	
148	PCIE1_TX0N	A17	O		PCIE1_TX0N	PCIE1 TX0 LANEN
150	PCIE1_TX0P	B17	O		PCIE1_TX0P	PCIE1 TX0 LANEP
152	GND		G	GND	Digital core Ground	
154	PCIE1_TX1N	B15	O		PCIE1_TX1N	PCIE1 TX1 LANEN
156	PCIE1_TX1P	A15	O		PCIE1_TX1P	PCIE1 TX1 LANEP
158	GND		G	GND	Digital core Ground	
160	PCIE0_REFCLK_N	C20	O		PCIE0_REFCLK_N	PCIE0 Clock LANEN
162	PCIE0_REFCLK_P	B20	O		PCIE0_REFCLK_P	PCIE0 Clock LANEP
164	GND		G	GND	Digital core Ground	
166	PCIE2/USB3-B_TX0N	A13	O		PCIE2/USB3-B_TX0N	USB3_B TX0 LANEN
168	PCIE2/USB3-B_TX0P	B13	O		PCIE2/USB3-B_TX0P	USB3_B TX0 LANEP
170	GND		G	GND	Digital core Ground	
172	PCIE5_TX0N	A7	O		PCIE5_TX0N	PCIE5 TX0 LANEN
174	PCIE5_TX0P	B7	O		PCIE5_TX0P	PCIE5 TX0 LANEP
176	GND		G	GND	Digital core Ground	
178	GPIO[3]_42 { GMAC2_RXDV / UART0_TXD / PCIEA_PERSTN / I2C0_SCL / PWM0 }	H34	O	1.8V	MOD_SLEEP	General purpose I/O3 42
180	GPIO[4]_81 { SSP0_TXD / SSPA0_CLK / R.UART4_RXD / CAN3_TXD / PCIEA_CLKREQN / I2C2_SDA }	AT34	I	3.3V	PCIEA_CLKREQn	PCIEA_CLKREQn
182	GPIO[4]_78 { R.SSPA0_TXD / SSPA2_TXD / UART8_CTS / PCIEE_CLKREQN / I2C1_SCL }	AP35	I	3.3V	PCIEE_CLKREQn	PCIEE_CLKREQn
184	PHY1_MDIO-		I/O		PHY1_MDIO-	PHY1 MDIO LANEN
186	PHY1_MDIO+		I/O		PHY1_MDIO+	PHY1 MDIO LANEP

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
188	PHY1_LED1/CFG_LDO0		O		PHY1_LED1/CFG_LDO0	PHY1_LED1/CFG_LDO0
190	PHY1_MDI1-		I/O		PHY1_MDI1-	PHY1 MDI1 LANEN
192	PHY1_MDI1+		I/O		PHY1_MDI1+	PHY1 MDI1 LANEP
194	PHY1_LED2/CFG_LDO1		O		PHY1_LED2/CFG_LDO1	PHY1_LED2/CFG_LDO1
196	PHY1_MDI2-		I/O		PHY1_MDI2-	PHY1 MDI2 LANEN
198	PHY1_MDI2+		I/O		PHY1_MDI2+	PHY1 MDI2 LANEP
200	GND		G	GND	Digital core Ground	
202	PHY1_MDI3-		I/O		PHY1_MDI3-	PHY1 MDI3 LANEN
204	PHY1_MDI3+		I/O		PHY1_MDI3+	PHY1 MDI3 LANEP
206	GPIO[5]_123 { MMC2_DAT[2] / SSPA1_FRM / UART6_RXD / R.UART0_RXD / PCIEB_PWRLED }	AY24	I/O	1.8V	MMC2_DAT2	General purpose I/O5 123
208	FAN_TACH(1V8)		I	1.8V	FAN_TACH	FAN TACH
210	PMIC_32K_OUT(3V3)		O	1.8V	PMIC_32K_OUT	PMIC 32KHz Clock
212	GPIO[3]_56 { GMAC2_INT_N / UART3_TXD / SSP0_FRM / R.UART3_TXD / PWM14 }	J37	I	1.8V	M2_ALERT	General purpose I/O3 56
214	GPIO[3]_69 { R.GMAC3_TX_D3 / SSPA4_CLK / ESPI0_D1 / E/DP1_HPD / DSI0_TE / SSP3_RXD }	H40	I	1.8V	FORCE_RECOVERY	General purpose Strp I/O3 69
216	GPIO[5]_125 { MMC2_DAT[0] / SSPA1_RXD / PCIED_WAKEN / E/DP1_HPD / PCIEB_EINTEG }	AU23	I/O	1.8V	MMC2_DAT0	General purpose I/O5 125
218	GPIO[5]_126 { MMC2_CMD / SSPA1_SYSClk / PCIED_CLKREQN / I2C5_SCL }	AV23	I/O	1.8V	MMC2_CMD	General purpose I/O5 126

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
220	GPIO[5]_101 { SSP3_SCLK / SSPA3_TXD / UART4_CTS / CAN4_RXD / PCIED_ATTN / MNCLK_OUT1 }	AU28	O	1.8V	I2S3_SDOUT	I2S3_SDOUT
222	GPIO[5]_102 { SSP3_FRM / SSPA3_RXD / UART4_RTS / CAN4_TXD / PCIED_PWRCTN / I2C1_SCL }	AV28	I	1.8V	I2S3_SDIN	I2S3_SDIN
224	GPIO[5]_100 { SSP3_RXD / SSPA3_FRM / UART4_RXD / R.CAN2_RXD / PCIED_PRSENT2N / CLK32K_OUT }	AT28	I	1.8V	I2S3_LRCK	I2S3_LRCK
226	GPIO[5]_99 { SSP3_TXD / SSPA3_CLK / UART4_TXD / R.CAN2_TXD / CLK_CAMCK4 }	AR28	O	1.8V	I2S3_CLK	I2S3_CLK
228	GPIO[5]_127 { MMC2_CLK / PCIED_PRSENT2N / I2C5_SDA / USB30-2_DRV }	AY23	I/O	1.8V	MMC2_CLK	General purpose I/O5 127
230	FAN_PWM(1V8)		O	1.8V	FAN_PWM	FAN PWM
232	GPIO[3]_54 { GMAC2_MDC / UART3_RTSN / SSP0_RXD / PCIEA_EINTEG / I2C1_SCL / PWM12 }	H38	O	1.8V	I2C1_SCL	I2C1_SCL
234	GPIO[3]_55 { GMAC2_MDIO / UART3_RXD / SSP0_SCLK / R.UART3_RXD / I2C1_SDA / PWM13 }	H37	I/O	1.8V	I2C1_SDA	I2C1_SDA
236	PWR_SSP_SCLK { GPIO[122] / UART0_TXD }	G40	O	1.8V	UART0_TXD_PMU	Debug UART0_TXD
238	PWR_SSP_FRM { GPIO[123] / UART0_RXD }	E36	I	1.8V	UART0_RXD_PMU	Debug UART0_RXD
240	GPIO[4]_84 { SSP0_FRM / SSPA0_RXD / UART9_TXD / USB30-1_DRV / PCIEA_PWRCTN / DSI0_TE }	AY34	I/O	3.3V	SLEEP/WAKE	General purpose I/O4 84

PIN	Pin definition	Pin NO.	Pad Type	IO power domain	Function for K3 CoM260 KIT	Default function description
242	GND		G	G	GND	Digital core Ground
244	GND					
246	GND					
248	GND					
250	GND					
252	VDD_IN(5V/20V)		P	5V/9~19V	VDD_IN	K3-CoM260 Power
254	VDD_IN(5V/20V)					
256	VDD_IN(5V/20V)					
258	VDD_IN(5V/20V)					
260	VDD_IN(5V/20V)					

4. Electrical, Mechanical, and Thermal Characteristics

4.1. Operating and Absolute Maximum Ratings

4.1.1. Recommended Operating Conditions

Symbol	Parameter	Minimum	Typical	Maximum	Unit
VDDDC	VDD_IN (MODULE_ID low)	4.75	5	5.25	V
	VDD_IN (MODULE_ID high)	12	-	20	V
	VCC_RTC	1.85	-	5.5	V

Note:

The current K3-CoM260 design supports high-voltage (12 V to 20 V) input power only. Carrier board designs must meet the following requirements:

1. Connect MODULE_ID (Pin 217) to 3.3 V through a pull-up resistor.
2. The MODULE_ID pull-up power supply must be established before VDD_IN; otherwise, the module may fail to boot normally.
3. For 5 V power mode support, please contact SpacemiT.

4.1.2. Absolute Maximum Ratings

Symbol	Parameter	Minimum	Maximum	Unit	Notes
VDDMAX	VDD_IN (MODULE_ID low)	-0.5	5.5	V	
	VDD_IN (MODULE_ID high)	-0.5	20.5	V	
	VCC_RTC	-0.3	7.0	V	
IDDMAX	VDD_IN I _{max}	-	5	A	
VM_PIN	Voltage applied to any powered I/O pin	-0.5	VDD + 0.2	V	When SYS_RESET* is high and the relevant I/O power rail is powered, the maximum voltage is VDD + 0.2. Before SYS_RESET* goes high, I/O pins must not be driven high (>0.5V). When SYS_RESET* is low, the maximum voltage that may be applied to any I/O pin is 0.5V.

Symbol	Parameter	Minimum	Maximum	Unit	Notes
TOP	Operating temperature	-20	70	°C	
TSTG	Storage temperature	TBD	TBD	°C	

5. Packaging

This section provides packaging information for the K3-CoM260.

TBD



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